Effect of coffee on endothelial function in healthy subjects: the role of caffeine.

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Coffee is one of the most widely used pharmacologically active beverages. The present study was designed to evaluate the acute effect of coffee ingestion on endothelial function in healthy individuals, and the potential role of caffeine. We studied 17 healthy young adults (28.9+/-3.0 years old; nine men), who were regular non-heavy coffee drinkers. The endothelial performance was estimated by endothelium-dependent FMD (flow-mediated dilatation) of the brachial artery before and 30, 60, 90 and 120 min after ingestion of a cup of caffeinated coffee (80 mg of caffeine) or the corresponding decaffeinated beverage (< 2 mg of caffeine) in two separate sessions, following a randomized single-blind cross-over design. There was no difference in baseline FMD values between the two sessions [7.78 compared with 7.07% after caffeinated and decaffeinated coffee respectively; P = NS (not significant)]. Caffeinated coffee led to a decline of FMD (7.78, 2.86, 2.12, 4.44 and 4.57% at baseline, 30, 60, 90 and 120 min respectively; $P < 0.001$). This adverse effect was focused at 30 ($P = 0.004$) and 60 min ($P < 0.001$). No significant effect on FMD was found with the decaffeinated coffee session (7.07, 6.24, 5.21, 7.41 and 5.20%; P = NS). The composite effect of the type of coffee consumed over time on FMD was significantly different ($P = 0.021$). In conclusion, coffee exerts an acute unfavourable effect on the endothelial function in healthy adults, lasting for at least 1 h after intake. This effect might be attributed to caffeine, given that decaffeinated coffee was not associated with any change in the endothelial performance.